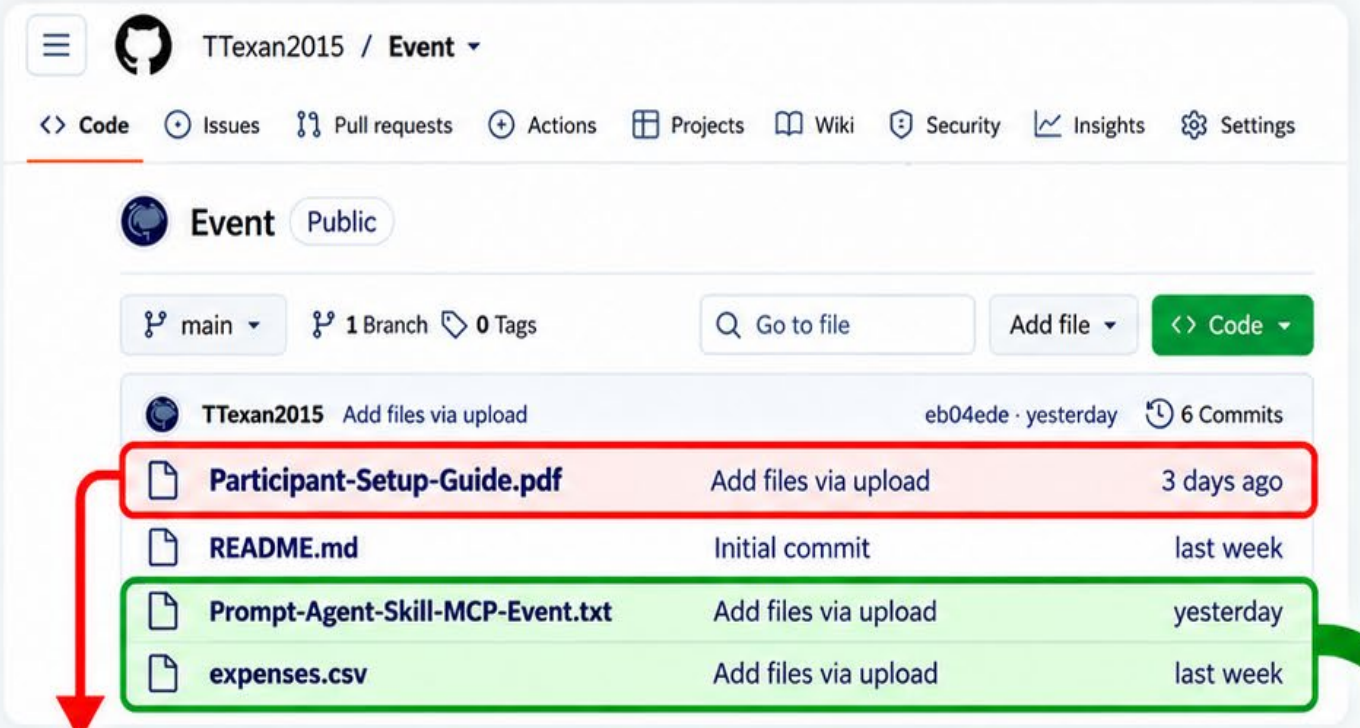
# Follow these steps to get the files for the event

## 1 Open this URL in your browser

**github.com/TTexan2015/Event**




## 2 Download the Participant Setup Guide

Click on **Participant-Setup-Guide.pdf** and download the file. Then follow the instructions in the guide.

## 3 Create a folder on your C drive

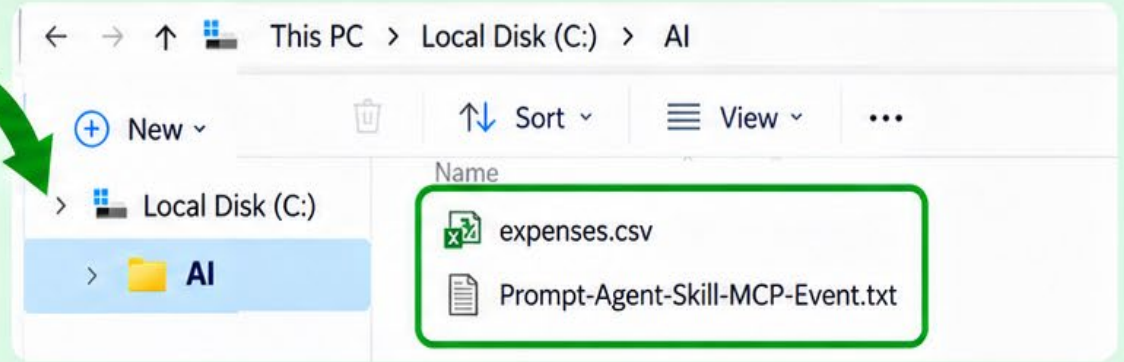
Open File Explorer, go to your C drive, and create a folder named AI (C:\AI).




## 4 Copy the two files into the C:\AI folder

From the GitHub repository, download the following two files and copy them into the C:\AI folder:

- **Prompt-Agent-Skill-MCP-Event.txt**
- **expenses.csv**



Microsoft

# Global AI Activate

# BUILD YOUR AI APPLICATION FROM SCRATCH

## AGENT-SKILL-MCP DESIGN PATTERN



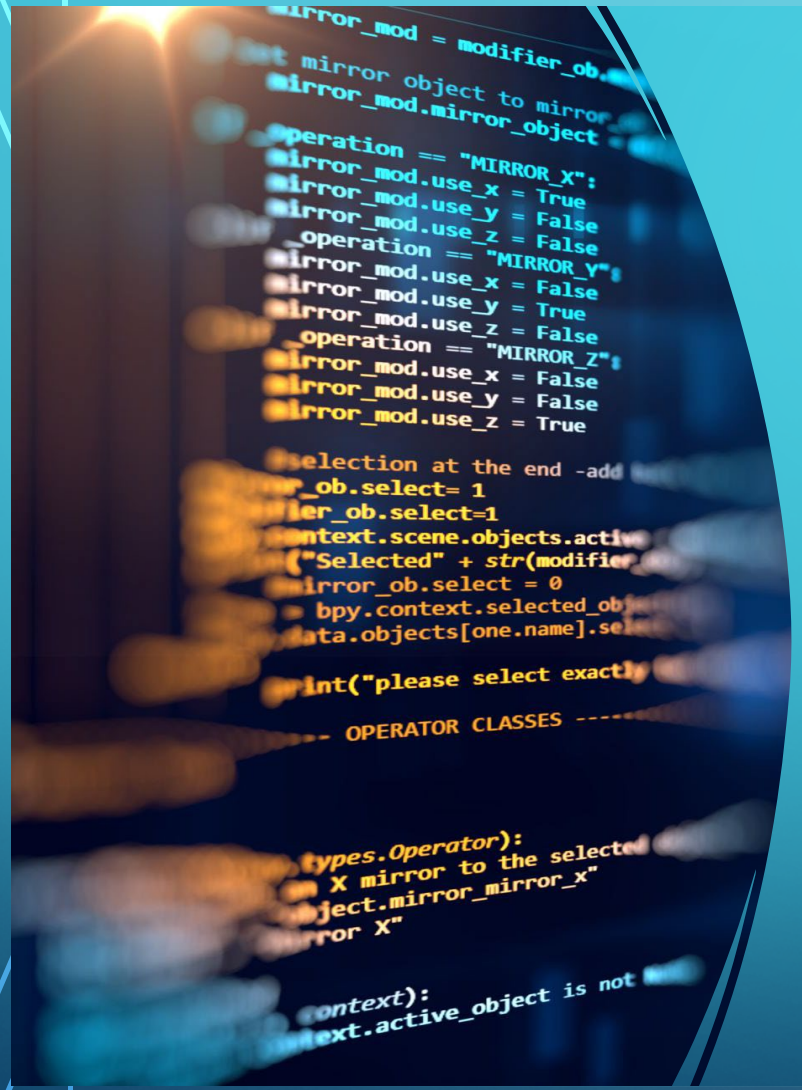

Tuesday
October 27

5:30 – 7:30 p.m.

Brought to you by **Microsoft**
Windows & Developer Azure Services

# Code of Conduct

✓ Be kind and inclusive: every level of experience is welcome, and every question is valued
✓ We follow the Microsoft and Global AI Community codes of conduct: no harassment or disrespect
✓ Anything make you uncomfortable? Tell any volunteer or site lead
✓ Protect your data: use only the sample data provided, no customer, personal or confidential data
✓ Never paste passwords, API keys or secrets into a prompt, a chat or a public repo
✓ Online: mics and cameras off · chat off · ask and upvote in Q&A · use reactions
✓ In the room: raise your hand for a lab coach · phones on silent
✓ Photos: tell our photographer if you'd rather not be in them · this session is not recorded




# BUILD YOUR AI APPLICATION FROM SCRATCH

## AGENT-SKILL-MCP DESIGN PATTERN

**NO PRIOR AI KNOWLEDGE NEEDED**

**Enamul Khaleque**
*Senior Escalation Engineer*
*AI Upskilling & Readiness co-lead*
Windows & Developer Azure Services
Microsoft Corp

**Christopher Corder**
*Senior Technical Advisor*
Windows & Developer Azure Services
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# Agenda

- The Agent-Skill-MCP design pattern
- Project Setup for AI Application
- Build Agent
- Add Skill
- Build MCP Server
- Connect Agent to MCP Server via Dynamic Looping
- Test the final AI Application

# AI Trivia

**1. The term “Artificial Intelligence (AI)” was formally introduced at this University?**

- **A. Harvard University**
- **B. MIT**
- **C. Dartmouth College**
- **D. Stanford University**

Answer: **C. Darthmouth College**

John McCarthy used the phrase in the 1955 proposal for the Dartmouth Summer Research Project. The workshop ran in Hanover, New Hampshire, in the summer of 1956. That is treated as the birth of the field.

# AI Trivia

**2. Who popularized the term “Agentic AI”?**

**A. Elon Musk**
**B. Andrew Ng**
**C. Donald Trump**
**D. LeBron James**

Answer: **B. Andrew Ng.**

Andrew Ng, co-founder of Google Brain, started using the term agentic AI in 2024, popularizing it in the AI research community.

# AI Trivia

**3. Which AI chat has the largest share today (based on Website visits)?**

**A. ChatGPT (OpenAI)**
**B. Grok (xAI)**
**C. Gemini (Google)**
**D. Claude (Anthropic)**

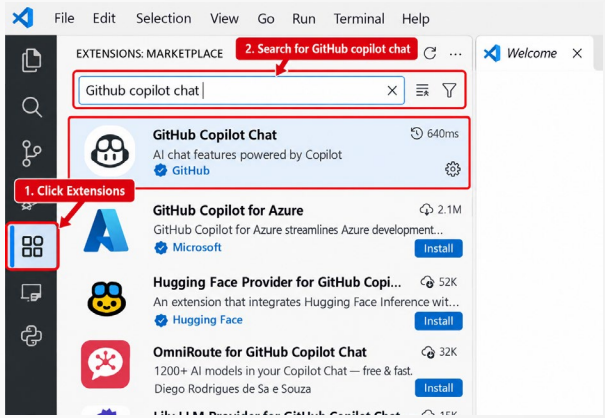
Answer: **A. ChatGPT (OpenAI)**

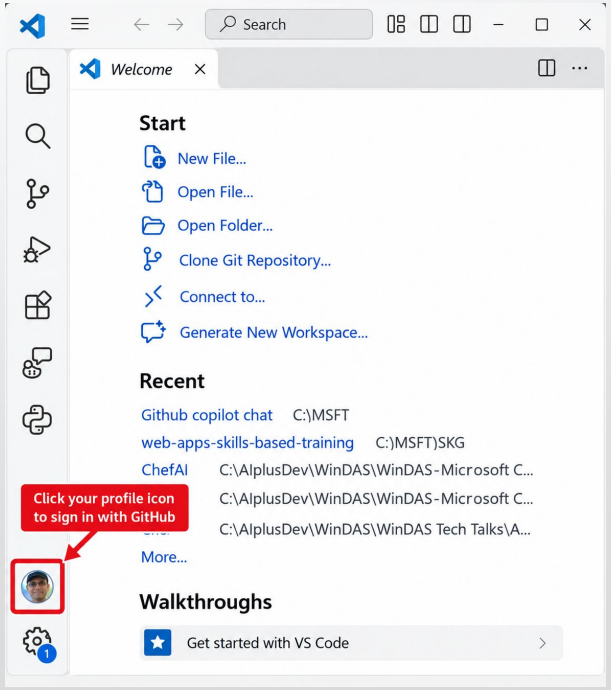
According to *First Page Sage*, as of October 1, 2026 report, “Top Generative AI Chatbots by Market Share: ChatGPT 52.1%, Gemini 26.9%, Claude 10.6%, Grok 3.4%

# Project Setup

- **Participant Setup Guide: https://github.com/TTexan2015/Event**

- ✓ **GitHub account: https://github.com/signup**
- ✓ **Visual Studio Code: https://code.visualstudio.com/download**
  - **Visual Studio Code includes GitHub Copilot Chat (if not included, then install from Extension)**
- ✓ **Sign-in to GitHub account within Visual Studio Code**
- ✓ **Install Python runtime: https://www.python.org/downloads/windows**
- ✓ **Set LLM Environment variables (for running the App, not used during building the app)**
  - **System Environment Variable: LLM_API_KEY, LLM_PROVIDER**
  - **LLM_PROVIDER value: OPENAI or ANTHROPIC or GROQ (If it is empty or not there, then OPENAI is chosen by default)**



# Are You Ready for Building Your AI Application?




## Microsoft AI Activate: Vibe Coding in Action

# The Kitchen Metaphor

## for Agents, Skills, MCP, and VS Code

*From ingredients to real results*




People Ideas Solutions

# Terminology



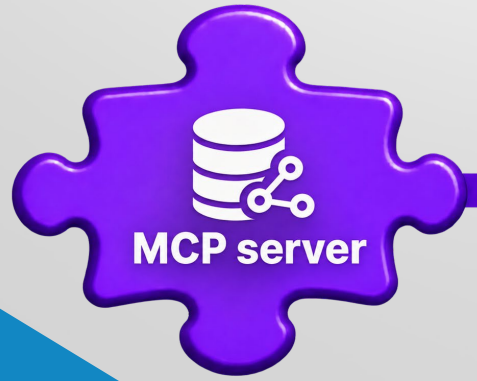

## Agent

- Receives the user request
- Understands what needs to be done
- Gets the work done and returns the result to the user



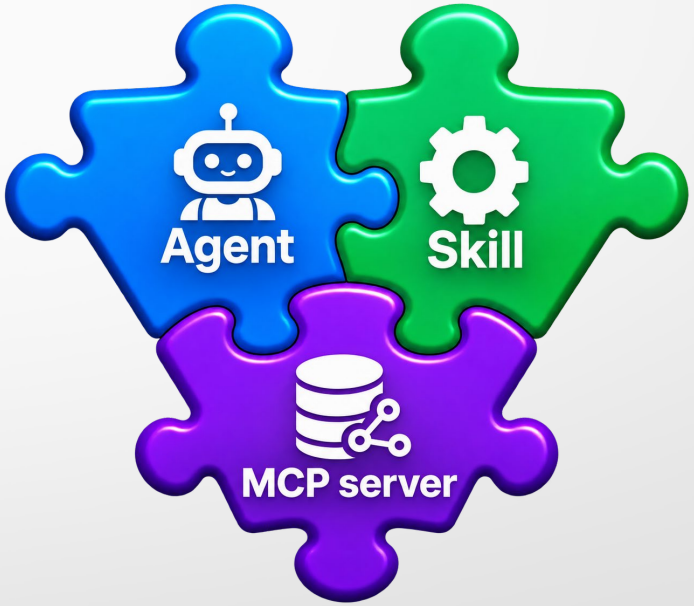

## Skill

- Helps the agent refine and improve its actions
- The agent taps into skills to improve its capabilities

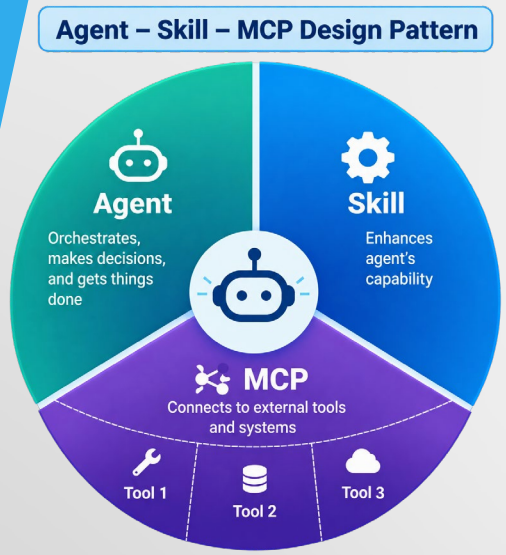


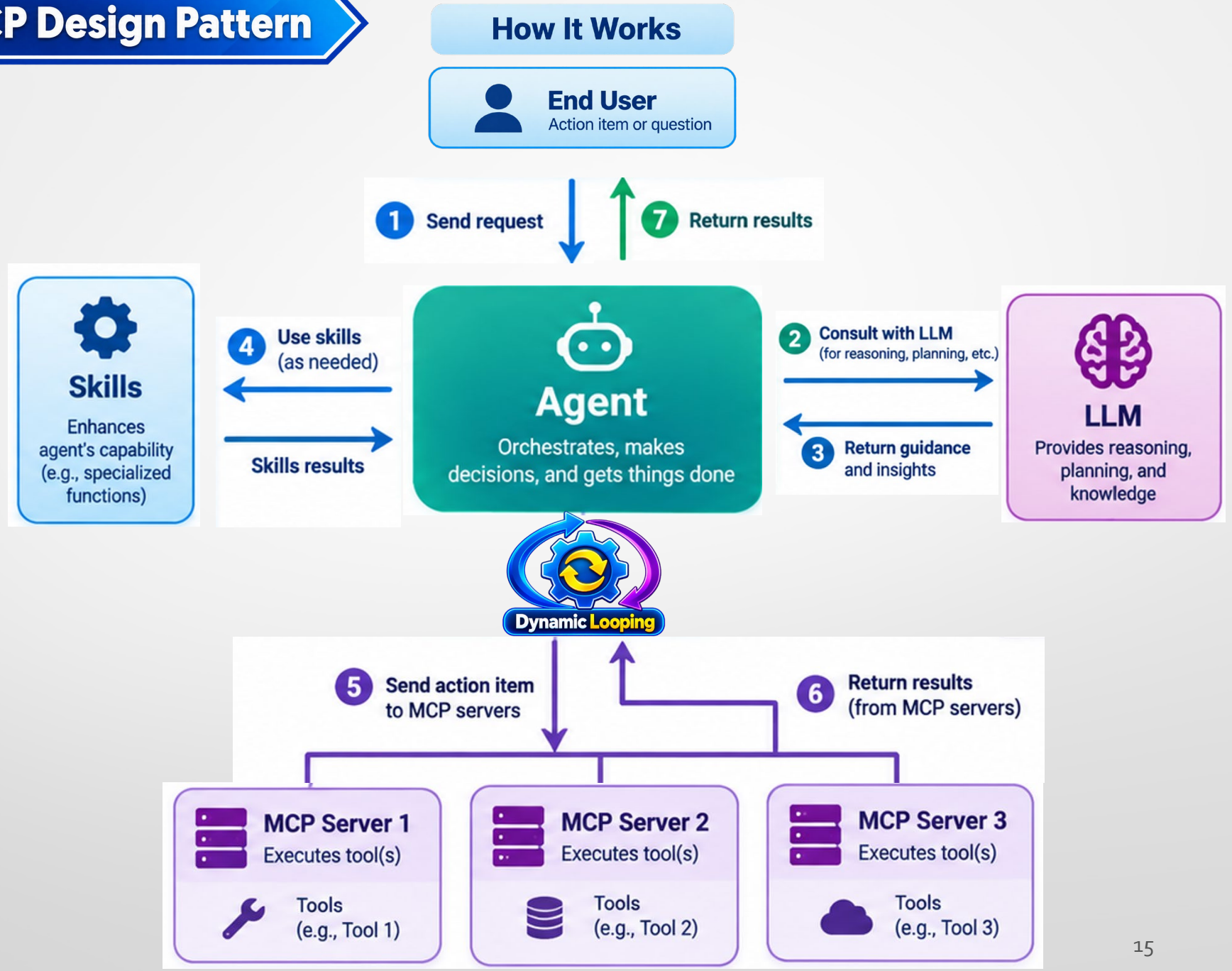

## MCP server

- MCP server (Model Context Protocol) is external to agent subsystem
- Provides features and tools that agent can use to do additional work

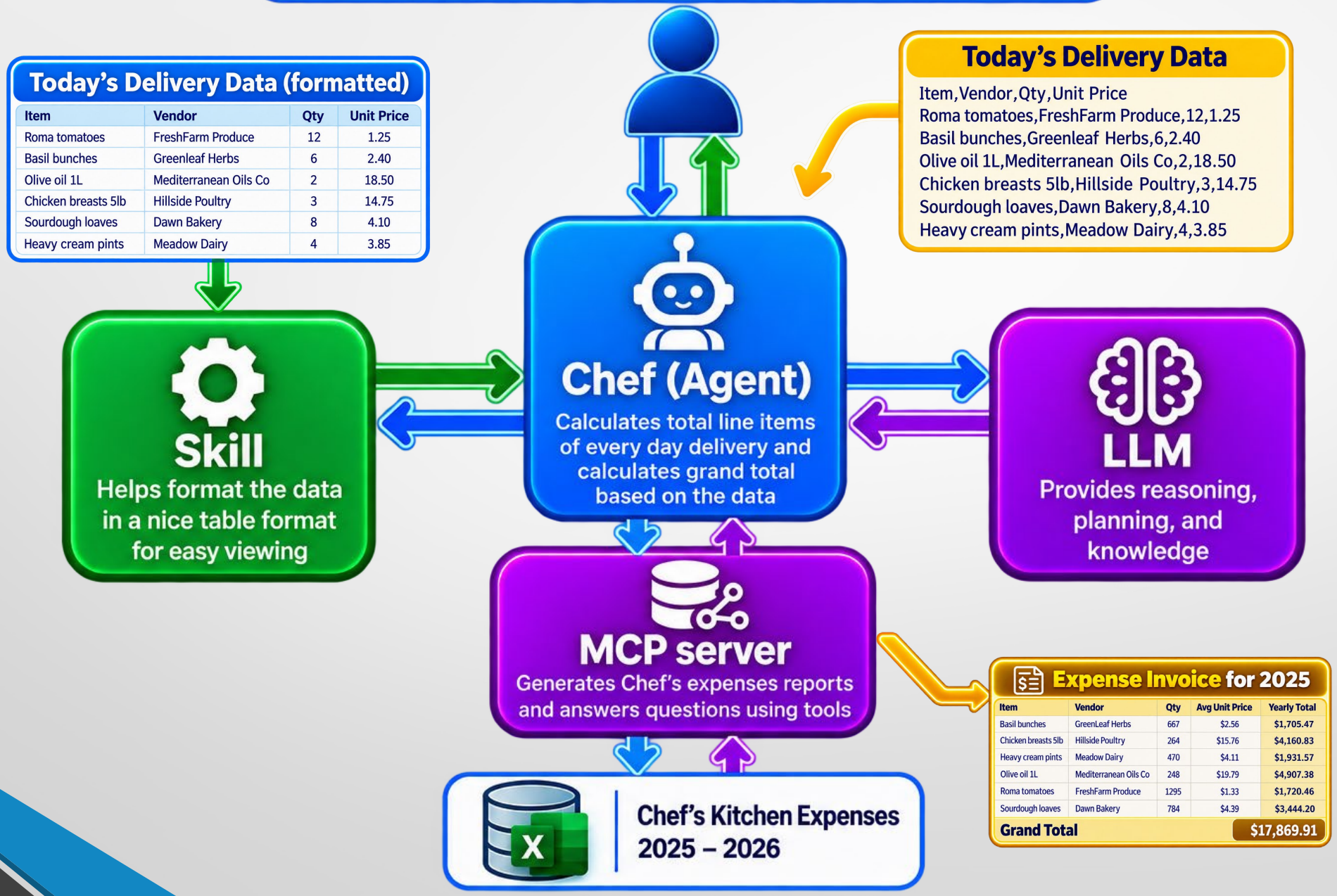


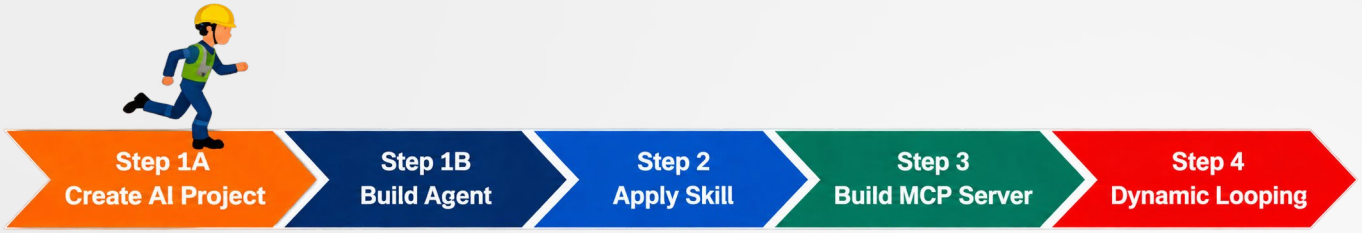
# Agent-Skill-MCP Design Pattern



# ChefAI : The Kitchen Expense Tracker






- Start in an empty VS Code folder
- Create only README.md, Agent.py, .env, and .gitignore
- Name the project from the folder name
- Leave the files empty
- No LLM call yet
- No extra folders



Step 1A Create AI Project | Step 1B Build Agent | Step 2 Apply Skill | Step 3 Build MCP Server | Step 4 Dynamic Looping

# Ready for the demo?

Download the demo files

**github.com/TTexan2015/Event**

1 Create a folder on your C drive

**C:\AI**

2 Copy these two files into C:\AI

C:\AI

- Prompt-Agent-Skill-MCP-Event.txt
- expenses.csv

Missing a file? Download it from the GitHub link.



Step 1A Create AI Project | Step 1B Build Agent | Step 2 Apply Skill | Step 3 Build MCP Server | Step 4 Dynamic Looping

**Today's Delivery Data**

Item,Vendor,Qty,Unit Price
Roma tomatoes,FreshFarm Produce,12,1.25
Basil bunches,Greenleaf Herbs,6,2.40
Olive oil 1L,Mediterranean Oils Co,2,18.50
Chicken breasts 5lb,Hillside Poultry,3,14.75
Sourdough loaves,Dawn Bakery,8,4.10
Heavy cream pints,Meadow Dairy,4,3.85

- Fill Agent.py with six hardcoded delivery items
- Python does the math. The LLM does not
- Print a delivery list with line total and today's total
- Read the API key from the environment (LLM_API_KEY)
- If the key is missing, skip the LLM and do not crash
- Call LLM to print two sentences under "Chef says:"




- Add a new SKILL.md: for formatting raw data into a clean table
- Keep the same items and the same totals
- Print the plain list first
- Load the skill and print the formatted table
- Numbers do not change. Only the layout changes
- “Chef says:” runs again on the formatted table



Step 1A Create AI Project | Step 1B Build Agent | Step 2 Apply Skill | Step 3 Build MCP Server | Step 4 Dynamic Looping

# What are Skills?

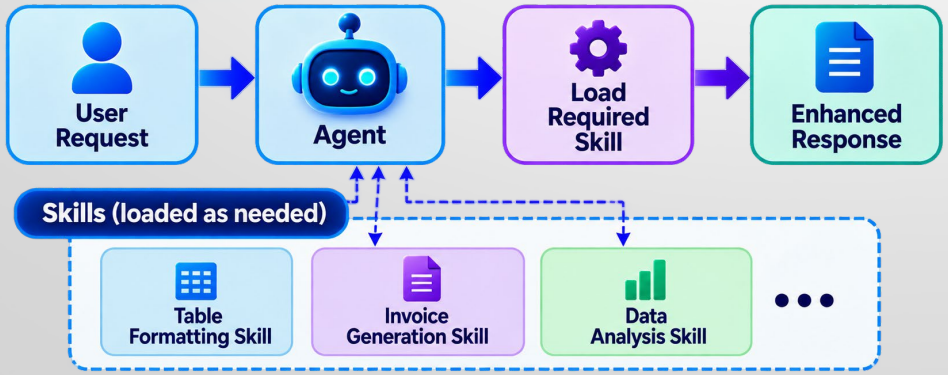



- **Reusable set of instructions, knowledge** that teaches an Agent how to perform a specific task.
- **Expands** Agent's capability.
- **Multiple skills** each dedicated to a specific capability.

**Project Structure (Example)**

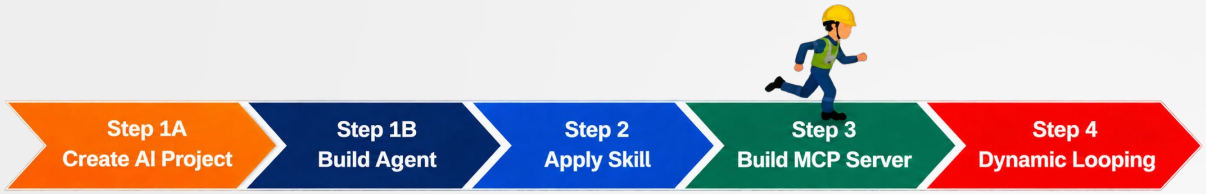
```
ChefAI/
  Agent.py
  README.md
  .env
  .gitignore
  skills/
    table_formatting/
      SKILL.md
    invoice_generation/
      SKILL.md
    data_analysis/
      SKILL.md
```

# How an Agent Uses Skills

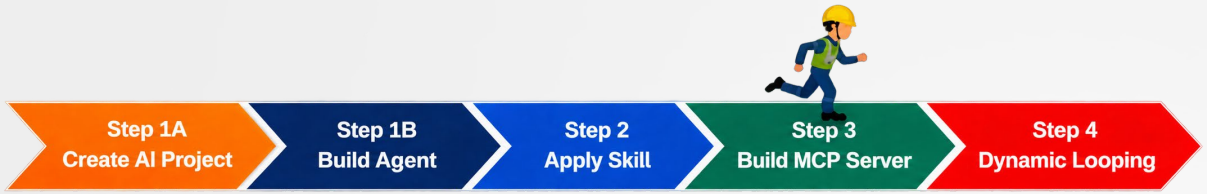



**Dynamic Selection**
Agent loads relevant skills as needed.

**Progressive Loading**
Agent first inspects skill descriptions, then loads details when needed.

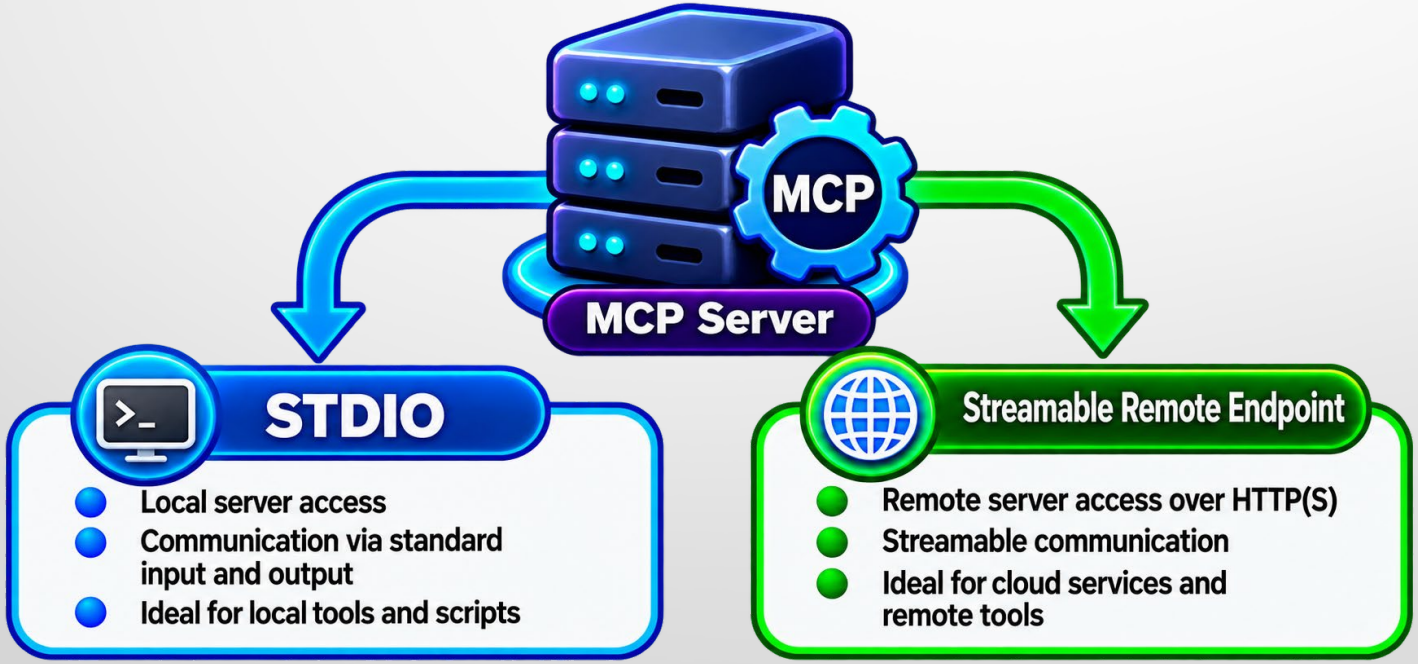


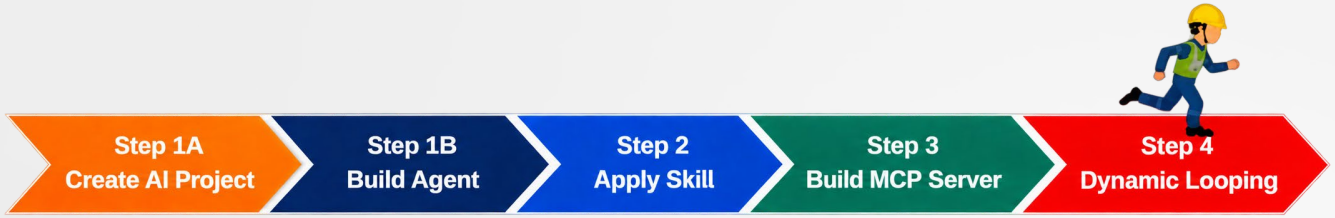

- New empty folder. Server only. No agent here
- Build a standalone Python MCP server from scratch
- Use the official MCP Python package named mcp
- Read the expense CSV datastore from parent/expenses.csv or from environment variable EXPENSES_CSV
- Expose following MCP tools: query_expenses, generate_invoice, and chef_summary
- MCP Server listening mode: local STDIO
- When a “demo” parameter is supplied, the MCP server writes three HTML invoices: latest delivery, year to date, last year




# MCP
## Model Context Protocol

- Standard protocol for AI models to access external tools and data
- Enables consistent and secure integration across different tools
- Defines how models discover, invoke and exchange information with tools






- Stay in the agent section
- Chose STDIO for communication over MCP
- Agent lists tools at runtime.
- Agent uses Dynamic Looping
- Agent asks the MCP Server to writes 3 “demo” invoices

❖ ==========

❖ Interactive mode -- choose an option

❖ ==========

❑ 1. Generate a custom invoice

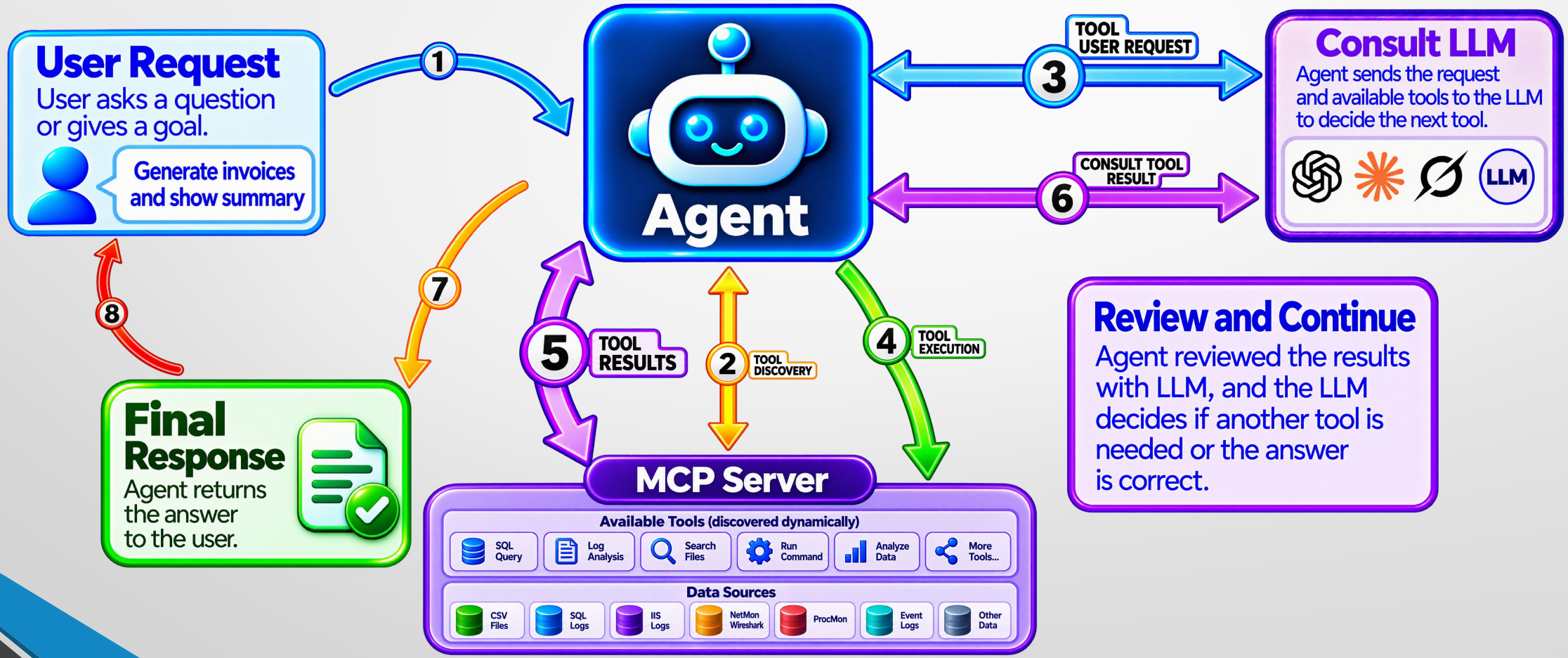
❑ 2. Ask a question about expenses

❑ 3. Exit

Step 1A Create AI Project | Step 1B Build Agent | Step 2 Apply Skill | Step 3 Build MCP Server | Step 4 Dynamic Looping

# Dynamic Looping in Agent-MCP Interaction

The agent dynamically discovers available tools from the MCP server, asks the LLM which tool to use, executes the tool, and repeats the process until the goal is met.




Question?